SOFTWARE REQUIREMENTS ENGINEERING (SRE) LABORATORY REPORT

Generating System and Context Diagrams

Proposed Scenario: University Hostel Management System

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1. Proposed Scenario

A university wants to develop a University Hostel Management System that allows students to apply for hostel accommodation online. Students can log in, view available rooms, submit room applications, view allocation/application status, check hostel fee information, and submit complaints. The Hostel Office manages hostel rooms, allocation rules, student allocations, and administrative information. The system communicates with a separate University Authentication System to verify user identities and with the University Finance System to verify hostel fees and payment status.

The purpose of this scenario is to provide a clear system boundary, identifiable external entities, meaningful information flows, and approximately 5–8 major internal modules, as required by the lab.

2. Student Analysis Worksheet

External Entity	Information Sent	Information Received	Notes / Assumptions
Student	Login credentials; room search request; hostel application; complaint details; fee/payment request	Authentication result; available rooms; application/allocation status; fee status; complaint status	Student is outside the system boundary because the student is a user of the software.
Hostel Office	Room details; room availability; allocation rules; management updates	Applications; allocation information; complaint information; administrative reports	Hostel Office is an external organizational role that operates the system.
University Authentication System	Authentication request and identity information	Identity verification/authentication result	Separate external software service; it is not part of the hostel system.
University Finance System	Payment/fee verification request	Payment result and fee status	Separate university financial service; availability of this service is an assumption.

3. Extracted Requirements

Requirement ID	Requirement
R1	Students must be able to authenticate and access the hostel system.
R2	Students must be able to view available hostel rooms.
R3	Students must be able to submit a hostel room application.
R4	Students must be able to view their application and allocation status.
R5	Students must be able to view hostel fee status and submit payment-related requests.
R6	Students must be able to submit and view hostel complaints.
R7	The Hostel Office must be able to manage rooms, availability, allocation rules, and hostel records.
R8	The system must communicate with the University

	Authentication System for identity verification.
R9	The system must communicate with the University Finance System for fee/payment verification.

4. Identify External Entities

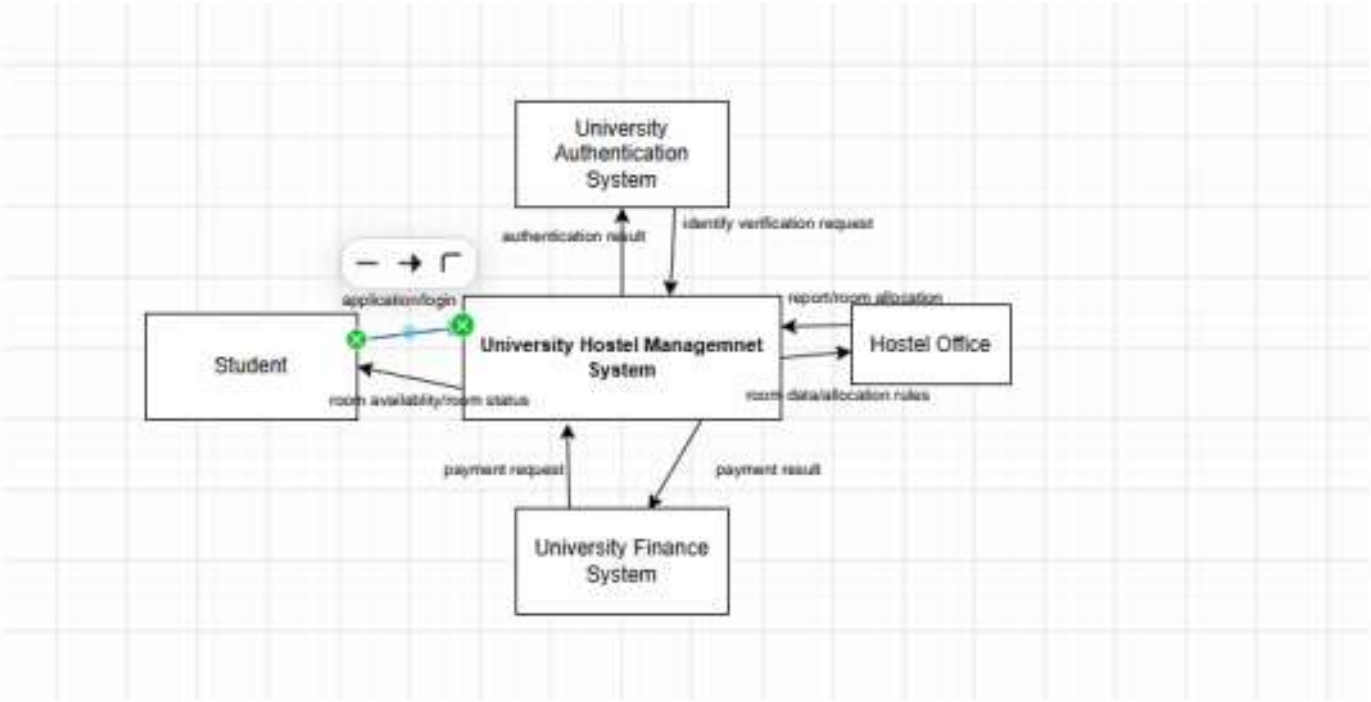
External Entity	Why External?	Major Interaction
Student	A person using the software, not an internal component.	Requests authentication, room information, applications, status, fees, and complaints.
Hostel Office	An external organizational role responsible for hostel administration.	Provides room/management information and receives applications, allocations, complaints, and reports.
University Authentication System	A separate external software service.	Verifies student identity.
University Finance System	A separate external software service.	Processes or verifies hostel fee/payment information.

5. Major Information/Data Flows

External Entity	Information Sent to System	Information Received	Related Requirements
Student	Login credentials; room search request; hostel application; complaint; fee request	Authentication result; room list; application status; allocation status; fee status; complaint status	R1–R6
Hostel Office	Room data; availability; allocation rules; management updates	Applications; allocations; complaints; reports	R7
University Authentication System	Authentication result / identity verification response	Authentication / identity verification request	R8
University Finance System	Payment result / fee verification response	Payment / fee verification request	R9

6. Context Diagram

The complete University Hostel Management System is treated as one central unit. External entities remain outside the boundary, and only major information flows are shown.



7. Context Diagram Explanation

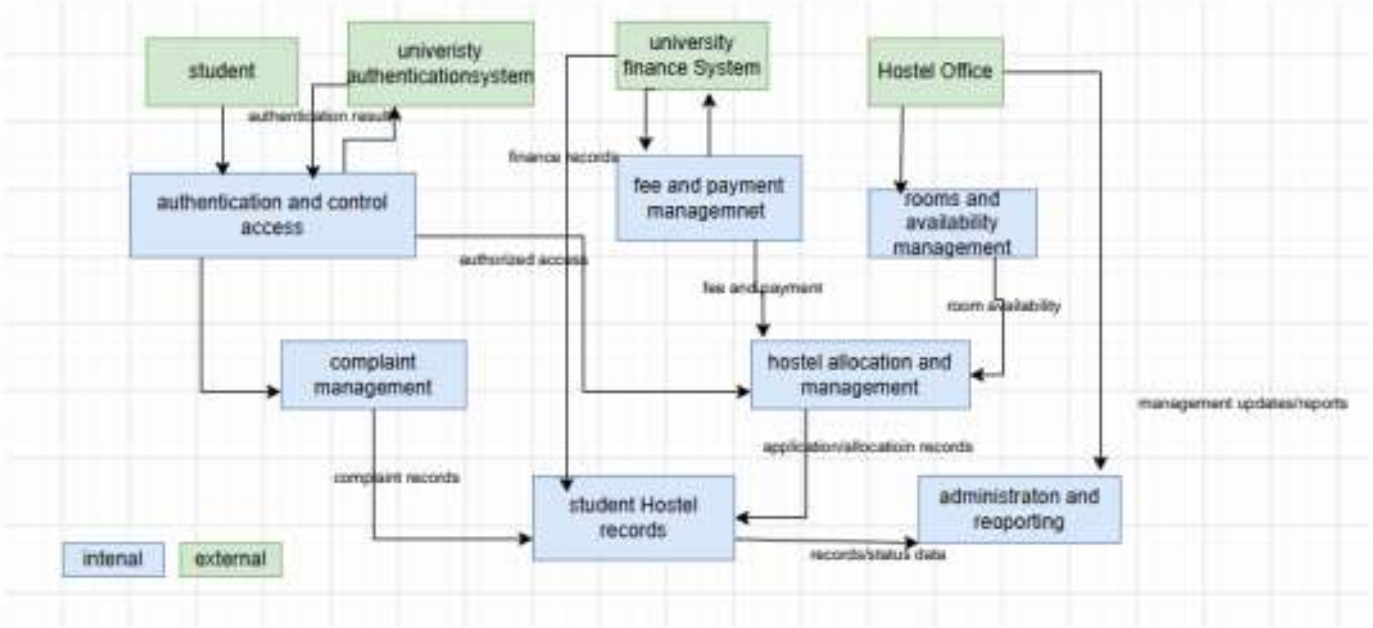
- The central box represents the complete University Hostel Management System.
- Student, Hostel Office, University Authentication System, and University Finance System are external entities because they are outside the target software boundary.
- The Student sends login credentials, room/application requests, fee requests, and complaint information and receives results/status information.
- The Hostel Office exchanges room, availability, allocation, complaint, and reporting information with the system.
- The Authentication System remains external because identity verification is performed by a separate service.
- The Finance System remains external because payment/fee verification is performed by a separate university service.
- Internal modules, databases, screens, and source-code functions are intentionally not shown in the context diagram.

8. Major Internal Modules

Internal Module	Responsibility
Authentication & Access Control	Handles login/access requests and communicates with the external authentication service.
Room & Availability Management	Maintains hostel room information and available-room status.
Hostel Application & Allocation	Processes hostel applications and supports room allocation according to rules.
Fee & Payment Management	Handles fee information and communicates with the finance system.
Complaint Management	Records and tracks hostel complaints submitted by students.
Student & Hostel Records	Maintains relevant student, application, allocation, fee, and complaint information.
Administration & Reporting	Supports Hostel Office management, updates, and reporting.

9. System Diagram

The system diagram opens the system boundary and shows the major internal responsibilities and their principal relationships. It remains conceptual rather than becoming a database or source-code design.



10. System Diagram Explanation

- Authentication & Access Control receives login/access requests and communicates with the external University Authentication System.
- Room & Availability Management provides current room availability to the application and allocation functions.
- Hostel Application & Allocation processes student applications and uses room availability and allocation rules.
- Fee & Payment Management communicates with the University Finance System and supplies fee status to the records area.
- Complaint Management records student complaints and updates student/hostel records.
- Student & Hostel Records maintains the information needed for status viewing, allocation history, fees, and complaints.
- Administration & Reporting supports Hostel Office management and uses the system's records to produce administrative information.

11. Context Diagram vs. System Diagram

Aspect	Context Diagram	System Diagram
Perspective	External / boundary view	Internal high-level view
System representation	One complete University Hostel Management System	Major internal modules
External entities	Main focus	Shown only where useful for interaction context
Internal modules	Not shown	Shown
Main purpose	Define system boundary and external interactions	Explain major internal organization
Level of detail	Very high level	More detailed, but still conceptual

12. Requirement-to-Diagram Traceability

Requirement	Context Diagram Evidence	System Diagram Evidence
R1 Authentication	Student ↔ System; System ↔ Authentication System	Authentication & Access Control
R2 View rooms	Student ↔ System	Room & Availability Management
R3 Apply for room	Student ↔ System	Hostel Application & Allocation
R4 View status	Student ↔ System	Student & Hostel Records
R5 Fees/payment	Student ↔ System; System ↔ Finance System	Fee & Payment Management
R6 Complaints	Student ↔ System	Complaint Management
R7 Hostel Office management	Hostel Office ↔ System	Administration & Reporting; Room & Availability; Application & Allocation
R8 Identity verification	System ↔ Authentication System	Authentication & Access Control
R9 Fee verification	System ↔ Finance System	Fee & Payment Management

13. Ambiguities and Assumptions

Ambiguity / Missing Detail	Assumption Made	Why It Matters
Who decides allocation priority?	The Hostel Office defines allocation rules.	Allocation logic must have a clear authority.
Does the system process money directly?	The system sends payment/verification requests to a separate Finance System.	This keeps the finance service outside the hostel system boundary.
How are complaints resolved?	The Hostel Office reviews and updates complaint status.	A status flow is needed for tracking.
Can students apply for multiple rooms?	The system allows one active hostel application per student unless the Hostel Office changes the rule.	Prevents ambiguous duplicate applications.
What happens if an external service is unavailable?	The system displays a clear service-unavailable message and does not falsely confirm authentication or payment.	Protects correctness of transactions.

14. Lab Review / Presentation Questions and Answers

Why did you classify this entity as external?
Student, Hostel Office, University Authentication System, and University Finance System are outside the software being modeled. The first two are users/organizational roles, while the latter two are separate external services.

Why is this component inside the system boundary?
The internal modules are responsibilities that the proposed University Hostel Management System itself performs, such as room management, applications, allocation, fees, complaints, records, and reporting.

What requirement supports this information flow?
Each flow is traced to R1–R9. For example, room availability is supported by R2, applications by R3, authentication by R1/R8, and fee verification by R5/R9.

What happens if the external service becomes unavailable?
The system should not confirm an operation that depends on the unavailable service. It should report that authentication or payment verification is temporarily unavailable and allow the user to retry.

Which requirement was ambiguous?
The original proposed scenario does not specify allocation priority, payment processing details, complaint-resolution rules, or duplicate-application rules. These were documented as assumptions.

What assumption did your group make?

We assumed that the Hostel Office defines allocation rules and that the Finance System handles or verifies payment information externally.

Why did you choose these internal modules?
They map directly to the main responsibilities in the scenario: authentication, room availability, applications/allocation, fees, complaints, records, and administration/reporting.

Could two modules be combined? Why?
Room & Availability Management and Application & Allocation could be combined in a very small system, but they are kept separate because room availability and application/allocation are distinct responsibilities.

Does the system diagram support all major functions in the scenario?
Yes. Authentication, room viewing, applications, allocation/status, fees, complaints, hostel administration, and external verification are represented.

Is anything in the diagram unsupported by the requirements?
The core modules are supported by the proposed requirements. Reporting is treated as an explicit administrative responsibility/assumption because the scenario requires management information but does not define detailed report types.

15. Additional Conceptual Questions

Why is Student outside the system boundary?
Because the student is a human user who interacts with the software; the student is not an internal software component.

Why is Hostel Application & Allocation inside the system?
Because processing hostel applications and allocations is a core function performed by the proposed target system.

Why is the University Authentication System external?
Because it is modeled as a separate university service responsible for identity verification.

Why is 'Room Application Request' a better flow name than simply 'Data'?
It identifies the actual information being exchanged and makes the diagram traceable to a requirement.

Why should a database not normally appear in this context diagram?
The context diagram is intended to show the system as one unit and its external interactions, not implementation details.

Why do we need a separate system diagram?
The context diagram defines the boundary, while the system diagram explains the major internal responsibilities that collectively implement the system.

Which requirement justifies Student & Hostel Records?
R4 and the status/history functions require the system to maintain relevant application, allocation, fee, complaint, and hostel information.

What would change if the university developed its own authentication module instead of using a separate authentication system?
Authentication would become an internal module, so it would no longer appear as an external entity or external flow in the context diagram.

16. Conclusion

The University Hostel Management System demonstrates how a textual problem statement can be converted into a clear requirements model. The context diagram establishes the system boundary and identifies the Student, Hostel Office, University Authentication System, and University Finance System as external entities. The system diagram then opens the boundary and decomposes the software into major internal responsibilities. The assumptions and traceability table help validate that diagram elements are supported by requirements rather than being added as unsupported implementation details.

17. Diagram File

The accompanying .drawio file contains editable diagrams that can be opened in diagrams.net (Draw.io). The Word report contains readable copies of both diagrams.